

Week 49 Four tower differences

Facilitator guide | Final student packets v2 | Prepared 3 October 2026 | Unpiloted

Mathematical overview

Simultaneous cyclic differences. With stations A,B,C,D in arrow order, the rule is $D(a,b,c,d) = (|a-b|, |b-c|, |c-d|, |d-a|)$. All four outputs use the same old state. The gap on an arrow is assigned to its tail station, including D to A. Updating a tower in place before computing the others is a different process.

For nonnegative integer starts, **every four-station run reaches zero**. The maximum never increases. Four rounds make all entries even; scaling commutes with the rule, so after $4k$ rounds every entry is divisible by 2^k . Once 2^k exceeds the original maximum, boundedness forces every entry to be zero. The maximum need not decrease at each move, and the total height is not conserved.

Limits and contrast. Three stations can enter a nonzero cycle. The argument uses integer heights; do not claim that arbitrary real starts satisfy this termination proof. Scaling by a positive integer preserves the number of moves to zero. A zero start takes zero moves; a nonzero constant start takes one.

Discovery by band. K-1 finds gaps physically and searches for short or long runs. Grades 2-3 tests predecessor obstructions and effects of changing station order. Grades 4-5 relates the parity shadow, scaling and a bound to a universal termination proof. Many successful trials are evidence, not the general proof.

Prepare and launch

Use two labeled old/new mats, number cards or erasable markers, pencils and optional cubes. Print at actual size. The paired working mats on pages 2 and 4 have 40 mm square stations. Four-station centers are 52 mm apart along a side; three-station centers span 52 mm horizontally and vertically. Small printed rings are reference diagrams, not cube-working spaces. Forty cubes per table suffice for two four-station states with heights at most 5; use number cards for the printed doubled start containing 8.

Readiness. K-1 matches cubes to find the unmatched part with adult reading and recording. Grades 2-3 needs small absolute differences. Grades 4-5 needs even/odd and reasoning about repeated scaling; the final proof is optional for children who are still productively exploring.

Launch, 5 minutes. First let children compare two towers. Build the printed old ring (1,4,2,0); keep it untouched while making new $A=3$ from A and B. Complete new $B=2$, $C=2$ and $D=1$ together. Only then swap old/new roles. Point to the wraparound arrow and keep labels fixed.

Flexible hour. 0-6 launch; 6-22 run starts; 22-25 stand and carry four gap cards to a new ring; 25-43 choose longevity, predecessors or three stations; 43-55 share findings or parity explanations; 55-60 reset. Keep one adult anchored to each of the three tables.

Key for the opening runs

Tuples below always use the printed station order. Each arrow counts one move, including the move that first produces all zeros.

Problem 1, all bands.

$(1,1,0,0) \rightarrow (0,1,0,1) \rightarrow (1,1,1,1) \rightarrow \text{zero}$: **3 moves**.

$(1,0,1,0) \rightarrow (1,1,1,1) \rightarrow \text{zero}$: **2 moves**.

$(2,1,0,1) \rightarrow (1,1,1,1) \rightarrow \text{zero}$: **2 moves**.

$(0,0,0,2) \rightarrow (0,0,2,2) \rightarrow (0,2,0,2) \rightarrow (2,2,2,2) \rightarrow \text{zero}$: **4 moves**, longest of the four printed starts.

Problem 2, all bands. The start $(0,1,2,4)$ is allowed both for K-1's 0-4 range and the older 0-5 range, and takes **7 moves**:

$(0,1,2,4) \rightarrow (1,1,2,4) \rightarrow (0,1,2,3) \rightarrow (1,1,1,3) \rightarrow$

$(0,0,2,2) \rightarrow (0,2,0,2) \rightarrow (2,2,2,2) \rightarrow \text{zero}$.

Independent exhaustive checks of all 625 and 1,296 permitted starts, respectively, give maximum 7. The prompt asks children to improve their attempts, not to reproduce that exhaustive proof.

Key for one and two moves

K-1 Problem 3. Exactly the positive constant starts (c,c,c,c) reach zero for the first time in one move. For example, $(1,1,1,1)$ and $(3,3,3,3)$. Every neighboring gap must be zero, so the whole ring must be constant. Do not count $(0,0,0,0)$, which takes zero moves.

K-1 Problem 4. Examples $(0,1,0,1)$ and $(0,1,2,1)$ both become $(1,1,1,1)$, then zero. In general, a two-move start has all four neighboring gaps equal to the same positive c . Positivity makes the first output nonzero; being constant makes the next output zero.

Older Problem 3, in printed reading order.

$(2,0,2,0)$: **possible**, from $(0,2,2,0)$.

$(1,1,1,0)$: **impossible**. Every output has an even total: modulo 2, each old height occurs in two gaps, so their sum is 0.

$(1,1,1,1)$: **possible**, from $(0,1,0,1)$.

$(0,0,0,2)$: **impossible**. The first three zero gaps force $a=b=c=d$, making the fourth gap zero too. Even total is necessary, not sufficient.

Key for changing the number of stations

K-1 Problem 5; older Problem 4.

Four stations: $(1,0,0,0) \rightarrow (1,0,0,1) \rightarrow (1,0,1,0) \rightarrow (1,1,1,1) \rightarrow \text{zero}$.

Three stations: $(1,0,0) \rightarrow (1,0,1) \rightarrow (1,1,0) \rightarrow (0,1,1) \rightarrow (1,0,1)$. The last three states form a cycle of period 3. The entire labeled state repeats, not merely its largest height or total. Determinism then forces this cycle to continue.

Key for station order and size

K-1 Problem 6; Grades 2-3 Problem 5. The order (0,1,2,4) lasts 7 moves; (0,1,4,2) lasts 4. All rotations and reversals of the first order also last 7. They are the eight longest labeled orders; the other sixteen permutations last 4. To check completeness, rotate 0 to A and identify reversal pairs: only three representatives remain, (0,1,2,4), (0,1,4,2), (0,2,1,4), with durations 7,4,4. The worksheet counts rotations as different whenever the A,B,C,D entries differ.

Grades 2-3 Problem 6; Grades 4-5 Problem 7. A gap between two numbers in $[0,M]$ is at most M . Applying this at every move shows that no later height exceeds the original maximum. Equality can persist: (0,1,2,4) becomes (1,1,2,4), with maximum 4 in both states. The rule does not say that each individual tower decreases.

Key for the parity and scaling investigations

Grades 4-5 Problem 5. Only EEEE is possible after four moves. Write $E=0$, $O=1$ and add modulo 2: $|a-b|$ has the same parity as $a+b$. From any (a,b,c,d) , the second parity row is $(a+c,b+d,a+c,b+d)$, the third is (s,s,s,s) with $s=a+b+c+d$, and the fourth is $(0,0,0,0)$. This symbolic argument covers all 16 starting parity patterns. Alternatively, children can organize and check all 16 cards.

Grades 4-5 Problem 6. The first printed start runs
 $(1,4,2,0) \rightarrow (3,2,2,1) \rightarrow (1,0,1,2) \rightarrow (1,1,1,1) \rightarrow \text{zero}$.

The doubled start runs

$(2,8,4,0) \rightarrow (6,4,4,2) \rightarrow (2,0,2,4) \rightarrow (2,2,2,2) \rightarrow \text{zero}$.

Each state is doubled, because $|2a-2b|=2|a-b|$. Tripling works by the same identity with 3. Both runs take 4 moves.

Grades 4-5 Problem 8. Let M be the original maximum. Problem 5 makes $D^4(h)$ even. If $D^{4k}(h)=2^k v$, then four further moves give $2^k D^4(v)$, whose entries are divisible by 2^{k+1} . Hence $D^{4k}(h)$ is divisible by 2^k for every k . Problem 7 bounds each entry by M . Choose k with $2^k > M$: the only nonnegative multiple in $[0,M]$ is zero. This proves finite termination. The bound $4k$ is convenient, not sharp.

Optional hints and sources

For an incorrect run, preserve both states and check one arrow at a time. For two-move starts, ask what the penultimate ring must look like. For parity, show one E/O conversion before asking for a classification. If the final proof stalls, ask separately what can grow and what divisibility accumulates; do not make the proof a prerequisite for enjoying the experiment.

Daniel Shapiro, *The Four-Number Game* (2005), pp. 1-3: [author's explanation and parity proof](#). The simultaneous labeled-mat presentation, exact final trajectories, permutation classification and bounded searches were independently verified for this packet. The general proof is separate from finite enumeration. Preparation and pacing are unpiloted proposals, with no physical-test claim.